

Variational Autoencoders

$$\max_{\theta} \log p_{\theta}(x) \rightarrow \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) || p(z))$$

① EM

$$1. \text{ Pdf of encoder: } q_{\phi}(z|x) \left(\int q_{\phi}(z|x) dz = 1 \right)$$

$$\Rightarrow \log p_{\theta}(x) \cdot 1 = \log p_{\theta}(x) \int q_{\phi}(z|x) dz$$

$$2. = \int q_{\phi}(z|x) \log p_{\theta}(x) dz$$

$$3. (P.16) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x)]$$

$$4. p_{\theta}(z|x) = p_{\theta}(z|x) \cdot p_{\theta}(x) \Rightarrow p_{\theta}(x) = \frac{p_{\theta}(z|x)}{p_{\theta}(z|x)}$$

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x)}{p_{\theta}(z|x)} \right]$$

$$5. = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x)}{p_{\theta}(z|x)} \cdot \frac{q_{\phi}(z|x)}{q_{\phi}(z|x)} \right]$$

$$6. = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x)}{q_{\phi}(z|x)} + \log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right]$$

7. by linearity of expectation

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x)}{q_{\phi}(z|x)} \right] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right]$$

8. 1st term: ELBO

2nd term: KL divergence (P.17) between q_{ϕ} and p_{θ}

$$9. \log p_{\theta}(x) = \text{ELBO} + \text{D}_{\text{KL}}(q_{\phi}(z|x) || p_{\theta}(z|x))$$

10. since $\text{KL} \geq 0$ (P.18)

$$\log p_{\theta}(x) \geq \text{ELBO}$$

$$11. \text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x)}{q_{\phi}(z|x)} \right]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(z|x) p_{\theta}(x)}{q_{\phi}(z|x)} \right]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(z|x)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x)}{q_{\phi}(z|x)} \right]$$

$$\therefore \text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(z|x)] - \text{D}_{\text{KL}}(q_{\phi}(z|x) || p(z))$$

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$$\max_{\theta} \log p_{\theta}(x) \rightarrow \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) || p(z))$$

② Jensen's inequality

$$1. \log p_{\theta}(x) = \log \int p_{\theta}(x|z) dz$$

$$2. = \log \int p_{\theta}(x|z) \cdot \frac{q_{\phi}(z|x)}{q_{\phi}(z|x)} dz$$

$$3. = \log \int q_{\phi}(z|x) \cdot \frac{p_{\theta}(x|z)}{q_{\phi}(z|x)} dz = \log \left(\mathbb{E}_{q_{\phi}(z|x)} \left[\frac{p_{\theta}(x|z)}{q_{\phi}(z|x)} \right] \right)$$

$$4. \text{ * Concave } f : f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

$$\text{since } \log \text{ is concave, } \log \left(\mathbb{E}_{q_{\phi}(z|x)} \left[\frac{p_{\theta}(x|z)}{q_{\phi}(z|x)} \right] \right) \geq \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x|z)}{q_{\phi}(z|x)} \right]$$

5. RHS is ELBO

$$6. = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x|z) \cdot p(z)}{q_{\phi}(z|x)} \right]$$

$$7. = \mathbb{E}_{q_{\phi}(z|x)} \left[\log p_{\theta}(x|z) \right] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right]$$

8. 2nd term : KL divergence (P.17)

$$\therefore \text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{D}_{\text{KL}}(q_{\phi}(z|x) || p(z))$$